

EDUCATION:

Brown University, Providence, RI | *Master of Science, Computer Science*

(Fall 2024 - May 2025)

Brown University, Providence, RI | *Bachelor of Science, Computer Science*

(Fall 2020 - May 2024)

WORK EXPERIENCE:

- **C++ Software Engineer, Dassault Systemes – Johnston, RI** (June 2025 - Present)
 - Shipped and benchmarked core execution layer optimizations, reducing perceived latency and improving reliability.
 - Built automated unit, integration, and E2E tests, capturing workflow regressions and boosting coverage by >50%.
 - Improved code modularity and performance to enable migration of a legacy workflow to modern architecture.
 - Leading design of internal AI productivity/triage tools, targeting reduction in time-to-root-cause for common failure modes.
- **Fullstack Software Engineer Intern, Onset Data Loggers - LI-COR – Bourne, MA** (May 2024 - August 2024)
 - Built full-cycle MQTT handler processing > 10k IoT messages daily, accelerating real-time device-Cloud communication.
 - Developed UI tool for firmware team to upload, edit, and delete JSONs for the Cloud, reducing manual setup time by 60%.
 - Prototyped a customizable mapping dashboard using Leaflet and GIS APIs, enabling user interaction with live weather data.
- **Head Teaching Assistant, Brown University CS – Providence, RI** (October 2023 - May 2024)
 - Managed a 100-student upper-level CS course; recruited and led 14 TAs, standardizing grading and student support.
 - Redesigned course projects to align with real-world UI/UX workflows, improving student engagement and enthusiasm.
 - Coordinated 20+ industry professionals for portfolio review sessions, increasing student exposure to industry standards.
- **Frontend Software Engineer Intern, Onset Data Loggers – Bourne, MA** (May 2023 - August 2023)
 - Delivered 2 production-ready features ahead of schedule for an environment data collection and management platform.
 - Collaborated with middleware engineers to improve API efficiency and reduce UI latency.
 - Completed multiple user stories per sprint in Agile workflows, expediting iteration cycles and release cadence.
- **Founder & Content Creator, Independent Art Business** (January 2020 - December 2023)
 - Built and scaled a digital art brand to 170k+ followers, reaching millions of views through data-driven content experimentation.
 - Produced and shipped physical products end-to-end, fulfilling 60+ orders and selling 300+ units for \$1.6k profit within months.
 - Partnered with Huion, XPPen, and Artisul in official collaborations to create multi-platform marketing and advertising content.

RECENT PROJECTS:

- **Ninja Mouse (Custom 3D Game Engine)**
 - Developed a C++ 3D game engine with component-based architecture, triangle-based collision detection, and AI pathfinding.
 - Abstracted engine components into 17 modules with OOP principles to allow scalability and robust per-game customization.
 - Integrated custom 3D assets with material loading support and environment shaders for enhanced visual fidelity.
 - Leveraged BVH and grid-based acceleration trees to logarithmically optimize environment creation and collision detection.
- **Origin Two (Custom 2D Game Engine)**
 - Developed Java-based 2D game engine with collision detection, AI pathfinding, animations, and dynamic level loading.
 - Implemented a precise, boundary-based physics system and custom UI framework for seamless player interaction.
 - Scaled engine to support 3+ dungeon crawler and platformer games alongside integrated sound and text generation.
- **Realtime Ocean Water Simulation**
 - Implemented a physically accurate ocean water simulation based on Jerry Tessendorf's paper on motion-picture oceans.
 - Built a fast fourier transform optimizer to allow real-time user manipulation based on wind speed and turbulence meters.
 - Introduced new features ocean foam, particle systems for spray, caustics, and custom shaders for reflection and refraction.
- **Realtime Graphics & Ray Tracing**
 - Created a real-time C++/OpenGL scene viewer with shape and camera manipulation and custom GLSL shaders.
 - Integrated ray tracing algorithm for realistic rendering using the Phong model, mirror reflections, and texture mapping.
 - Programmed custom shaders with framebuffer objects to achieve stylized post-processing effects.
- **3D Modeling & Animation**
 - Animated and imported game-ready and cyclable attack sequences with a custom 3D rig and character into Unreal.
 - Designed, modeled, rigged, textured, and animated 2 original characters with custom shaders for an animated short.
 - Created adaptable shader and texture customization sliders, supporting rapid creation of varying environments.
 - Led storyboarding, character design, and character modeling for a 3 minute animated short produced within 1 month.

SKILLS:

- **Computer Languages:** C++, Python, Java, React.js, Vue.js, Node.js, JS, TS, Shell, HTML/CSS, OpenGL, TensorFlow, Scipy
- **Tools:** Unreal, Maya, Blender, Gaea, Adobe, Git, Docker, Figma,
- **Skills:** Atlassian/Jira, Agile, CI/CD, Scrum, Kanban, Project Management